INCH-POUND
MIL-M-24519C(NAVY)
8 October 1987
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15 August 1983
(See 6.6)

MILITARY SPECIFICATION

MOLDING PLASTICS, ELECTRICAL, THERMOPLASTIC

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the basic properties of thermoplastic molding compounds used to mold electric and electronic component parts.

1.2 Classification. The thermoplastic molding compounds governed by this specification shall be of the following types, as specified (see 6.2.1).

Types	Description
PT-F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, flame resistant
GPT-7.5F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 7.5 percent glass reinforced, flame resistant
GPT-15F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate compound, 15 percent glass reinforced, flame resistant
GPT-20F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate compound, 20 percent glass reinforced, flame resistant

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Types	Description
GPT-30F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 30 percent glass reinforced, flame resistant
GAT-22F	Thermoplastic, polyarylether compound, 22 percent glass reinforced, flame resistant
GET-30F	Thermoplastic polyester (polyethylene terephthalate) compound, 30 percent glass reinforced, flame resistant
GMT-40F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 40 percent glass and mineral reinforced, flame resistant
GLT-10F	Thermoplastic polyetherimide compound, 10 percent glass reinforced, flame resistant
GLT-20F	Thermoplastic polyetherimide compound, 20 percent glass reinforced, flame resistant
GLT-30F	Thermoplastic polyetherimide compound, 30 percent glass reinforced, flame resistant
GST-40F	Thermoplastic polyphenylene sulfide compound, 40 percent glass reinforced, flame resistant
MLT-20F	Mineral plastic polytherimide compound, 20 percent mineral filled, flame resistant
GLCP-30F	Thermoplastic polyester, 30 percent glass reinforced liquid crystalline
GLCP-50F	Thermoplastic polyester, 50 percent glass reinforced liquid crystalline

2. APPLICABLE DOCUMENTS

2.1 Government publication. The following Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NATIONAL TECHNICAL INFORMATION SERVICE
AD297457 - Procedure and Analytical Method for Determining Toxic Gases Produced by Synthetic Compounds.

(Application for copies should be addressed to Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.)

(Copies of other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 149 - Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies. (DoD adopted)

D 150 - Standard Test Methods for A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials. (DoD adopted)

D 229 - Standard Method of Testing Rigid Sheet and Plate Materials Used for Electrical Insulation. (DoD adopted)

D 256 - Standard Test Methods for Impact Resistance of Plastics and Electrical Insulating Materials. (DoD adopted)

D 495 - Standard Test Method for High-Voltage, Low-Current, Dry Arc Resistance of Solid Electrical Insulation. (DoD adopted)

D 570 - Standard Test Method for Water Absorption of Plastics. (DoD adopted)

D 638 - Standard Test Method for Tensile Properties of Plastics. (DoD adopted)

D 648 - Standard Test Method for Deflection Temperature of Plastics Under Flexural Load. (DoD adopted)

D 695 - Standard Test Method for Compressive Properties of Rigid Plastics. (DoD adopted)

D 790 - Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials. (DoD adopted)

D 3220 - Standard Specification for Reinforced Thermoplastic Polyester Molding and Extrusion Materials. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORY INCORPORATED (UL)

UL 94 - Tests for Flammability of Plastic Materials for Parts in Devices and Appliances.

(Application for copies should be addressed to the Underwriters Laboratory Incorporated, 333 Pfingsten Road, Northbrook, IL 60062.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

Publication 112 (3rd edition) - Method for Determining the Comparative and the Proof Tracking Indices of Solid Insulating Materials Under Moist Conditions.

(Application for copies should be addressed to United States of America Standards Institute, Sales Department, 10 East 40th Street, New York, NY 10016.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Thermoplastic molding compounds furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.1.1 Qualification for compounds and colors. Contractors shall obtain qualification approval for each compound. Material qualified for one color (natural) will not be qualified for all other colors. The material (virgin) shall have moisture content no more than 1 percent maximum, to prevent from excessive warpage. A sealed bag shall be used for shipping and storage. Bags shall be be kept sealed until its use.

3.2 Materials.

3.2.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Material shall be free from foreign substances and dry to 1 percent moisture maximum content. None of the above shall be interpreted to mean that the use of used products is allowed under this specification unless otherwise specifically specified.

3.2.2 Uniformity. Molding compound of the same type from one contractor shall be uniform in texture, in color (see 3.2.3), and in the properties specified herein as determined by the quality conformance inspection specified in 4.4.

3.2.3 Color. Unless otherwise specified (see 6.2.1), materials shall be provided in natural color. The color shall be designated for QPL testing (see 6.4). Natural-colored compounds are those in which no coloring matter of any sort has been added. In evaluating uniformity of color, consideration shall be given to the fact that variation in color of raw materials may be reflected in the color of the natural molding compound. Coloring matters reducing the electrical properties below the specified limits shall not be used.

3.2.4 Material property values. Standard specimens of the compounds shall conform to the property values specified in table I.

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TABLE I. Property values for qualification tests of thermoplastic molding compounds.

Property to be tested	Value required for each type of compound															Units
	Type PT-F	Type GPT-7.5F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	Type MLT-20F	Type GLCP-30F	Type GLCP-50F	
Arc resistance	50	55	65	60	70	80	70	70	85	85	85	30	85	130	130	Seconds
Compressive strength, 10 percent deflection	12,000	12,500	13,500	14,000	16,200	13,500	17,500	16,200	19,000	20,000	22,000	20,000	20,000	11,000	11,000	lb/in^2
Dielectric constant:																
At 100 Hz	3.6	3.8	3.9	3.9	4.1	3.8	4.1	4.1	3.4	3.4	3.6	4.3	3.5	5.0	5.5	---
At 10^6 Hz	3.4	3.6	3.7	3.7	4.0	3.8	4.0	4.0	3.5	3.5	3.6	4.2	3.5	4.8	5.2	---
Dissipation factor:																
100 Hz	0.005	0.0050	0.005	0.005	0.010	0.003	0.010	0.005	0.007	0.012	0.004	0.002	0.004	0.04	0.04	---
10^6 Hz	0.03	0.0300	0.03	0.03	0.03	0.02	0.03	0.03	0.009	0.012	0.009	0.007	0.009	0.04	0.04	---
Dielectric strength:																
Short-time test	350	360	370	375	400	400	400	400	450	450	450	320	450	480	480	Volt/mil
Short-time test	325	330	340	350	375	375	375	375	430	430	430	300	430	480	480	
Dielectric breakdown:																
Short-time test	43	45	46	46	50	50	50	50	56	56	56	40	56	56	56	Kilovolt
Short-time test	40	41	42	42	46	46	46	46	54	54	54	37	54	56	56	
Flame resistance:																
Ignition time	50	55	55	50	55	55	50	55	65	65	65	50	65	100	100	Seconds
Burning time	35	50	50	50	20	50	20	20	20	20	20	20	20	50	50	
Flexural strength	13,000	15,000	19,000	20,000	24,000	20,000	24,000	24,000	26,600	28,500	31,400	22,000	22,500	25,000	28,000	lb/in^2
Flammability	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	---
Heat-deflection temperature	160	270	360	375	385	300	400	385	400	400	400	475	410	425	425	°F
Impact strength	0.6	0.6	0.7	0.8	1.1	0.8	1.1	1.1	0.9	1.0	1.1	1.1	0.6	1.3	0.8	ft-lb/in

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TABLE I. Property values for qualification tests of thermoplastic molding compounds. - Continued

Property to be tested	Value required for each type of compound															Units
	Type PT-F	Type GPT-7.5F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	Type MLT-20F	Type GLCP-30F	Type GLCP-50F	
Tensile strength	8,000	8,400	11,500	12,000	15,000	12,000	15,000	15,000	16,000	18,700	23,000	16,000	13,000	18,000	21,000	lb/in^2
Water absorption	0.40	0.35	0.35	0.30	0.30	0.40	0.30	0.30	0.69	0.60	0.55	0.08	0.60	0.08	0.08	Percent
Heat resistance	50	50	50	50	50	70	50	50	85	85	85	95	95	85	85	Percent
Dimensional stability (high temperature)	0.2	0.2	0.2	0.2	0.2	.12	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	Percent
Volume resistance	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1x10^{11}$	$1.0x10^9$	$1.0x10^9$	$1x10^{11}$	$1x10^{11}$	$1x10^{10}$	$1.0x10^{12}$	$1x10^{11}$	$1x10^{10}$	$1x10^{10}$	Ohm
Surface resistance	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$3.5x10^{10}$	$1.0x10^9$	$1.0x10^9$	$1x10^{10}$	$1x10^{10}$	$1x10^{10}$	$1.0x10^{11}$	$1x10^{10}$	$1x10^9$	$1x10^9$	Ohm
Tracking index	180	180	180	180	180	125	180	180	135	135	135	150	140	120	140	Volt
Toxicity when heated:																
Carbon dioxide	15,000	15,000	15,000	15,000	15,000	10,000	15,000	15,000	5,000	5,000	5,000	1,500	5,000	15,000	15,000	
Carbon monoxide	1,000	1,000	1,000	1,000	1,000	500	1,000	1,000	250	250	250	200	250	1,000	1,000	Parts
Ammonia	2,500	2,500	2,500	2,500	2,500	50	2,500	2,500	20	20	20	100	20	500	500	per
Aldehydes as H.CHO	50	50	50	50	50	50	50	50	20	20	20	20	20	50	50	million
Cyanide as HCN	60	60	60	60	60	50	60	60	20	20	20	0	20	50	50	(p/m)
Oxides of nitrogen as NO_2	100	100	100	100	100	60	100	100	50	50	50	25	50	100	100	
Hydrogen chloride	100	100	100	100	100	50	100	100	50	50	50	0	50	50	50	

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Naval Sea Systems Command (NAVSEA). Qualification inspection shall consist of the tests specified in tables I and II.

4.3.1 Specimens for qualification inspection. Tests shall be performed on standard test specimens molded by the manufacturer of each compound for which qualification is required. The number of test specimens to be tested shall be as specified in tables II and III.

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TABLE II. Sampling and conditioning for qualification tests.

Property to be tested	Test method		Specimens		Conditioning procedure (see 4.5.1.5)	Unit of value
	ASTM	Modified by	Form and dimensions	Number tested		
Arc resistance	D 495	-----	4-inch disk, 1/8 inch thick	3	A	Seconds (minimum average)
Compressive strength, 10 percent deflection	D 695	-----	1 by 1/2 by 1/2 inch	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Dielectric constant: At 100 Hz At 10^6 Hz	D 150	-----	4-inch disk, 1/8 inch thick	3 3 3 3	E-48/50+des E-48/50+D-24/23 E-48/50+des E-48/50+D-24/23	Maximum average
Dissipation factor: At 100 Hz At 10^6 Hz	D 150	-----	4-inch disk, 1/8 inch thick	3 3 3 3	E-48/50+des E-48/50+D-24/23 E-48/50+des E-48/50+D-24/23	Maximum average
Dielectric strength: Short-time test Short-time test	D 149	4.5.2.1 & 4.5.2.1.1	4-inch disk, 1/8 inch thick 4-inch disk, 1/8 inch thick	3 3	E-48/50+C-96/23/50 E-48/50+D-48/50	Volt per mil (minimum average)
Dielectric breakdown: Short-time test Short-time test	D 149	4.5.2.1.2	4-inch disk, 1/8 inch thick	3 3	E-48/50+C-96/23/50 E-48/50+D-48/50	Kilovolt (minimum average)
Flame resistance: Ignition time Burning time	D 229	4.5.2.9	5-inch bar, 1/2 by 1/2 inch	5	A	Seconds (minimum average) Seconds (maximum average)
Flexural strength	D 790	4.5.2.2	5-inch bar, 1/4 by 1/2 inch	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Flammability	---	4.5.2.10	5-inch bar, 1/2 x thickness	5	(see 4.5.2.10)	(see 4.5.2.10)
Heat-deflection temperature	D 648	4.5.2.3	5-inch bar, 1/2 by 1/4 inch	3	A	Degrees Fahrenheit (minimum average)
Impact strength	D 256	----	In accordance with ASTM D 256 and ASTM D 3220 except thickness of SP	5	E-48/50+C-96/23/50	ft-lb/in notch (minimum average)

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TABLE II. Sampling and conditioning for qualification tests. - Continued

Property to be tested	Test method		Specimens		Conditioning procedure (see 4.5.1.5)	Unit of value
	ASTM	Modified by	Form and dimensions	Number tested		
Tensile strength	D 638	----	In accordance with ASTM D 638	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Water absorption	D 570	4.5.2.4	2-inch disk, 1/8 inch thick	3	E-24/100+des+D -48/50	Percent (maximum average)
Heat resistance	D 790	4.5.2.5	5-inch bar, 1/4 by 1/2 inch	5	E-48/50+C-96/23/50 +E-1/200	Percent flexural strength retained (minimum)
Dimensional stability (high temperature)	----	4.5.2.6	5-inch bar, 1/2 by 1/2 inch	5	(see 4.5.2.6)	Percent (maximum average)
Volume resistance	----	4.5.2.7	4-inch disk, 1/8 inch thick	5	(see 4.5.2.7)	Ohm (minimum individual)
Surface resistance	----	4.5.2.7	4-inch disk, 1/8 inch thick	5	(see 4.5.2.7)	Ohm (minimum individual)
Tracking index	----	4.5.2.11	4-inch disk, 1/8 inch thick	3	(see 4.5.2.11)	Volt
Toxicity when heated: Carbon dioxide Carbon monoxide Ammonia Aldehydes as H.CHO Cyanide as HCN Oxides of nitrogen as No_2 Hydrogen chloride	----	4.5.2.8	5-inch bar, 1/2 by 1/2 inch	4	A	p/m parts per million

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TABLE III. Standard test specimens for qualification tests.

Specimen designation	Specimen description	Number of specimens for each type	Inches 1/	mm
1	4-inch disk by 1/8 inch	52	1/8 (.125)	3.18
2	1 by 1/2 by 1/2 inch	5	1/4 (.250)	6.35
3	2-inch disk by 1/8 inch	3	1/2 (.500)	12.70
4	5 by 1/2 by 1/2 inch	14	1 (1.00)	25.40
5	5 by 1/2 by 1/4 inch	13	2 (2.00)	50.80
6	2-1/2 by 1/2 by 1/2 inches (ASTM D 256)	5	2-1/2 (2.50)	63.50
7	ASTM D 638	5	4 (4.00)	101.60
8	5 by 1/2 inches by thickness	5	5 (5.00)	127.00

1/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 millimeter (mm).

4.3.2 Preparation. Specimens shall be prepared for tests as specified in 4.5.1.

4.4 Quality conformance inspection. Quality conformance shall be as specified in table IV. Tests shall be performed at the manufacturer's plant on standard specimens molded by the manufacturer from each batch of compound to be provided to molders for production of molded parts under this specification.

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TABLE IV. Property values for quality conformance inspection of thermoplastic molding compounds.

Property to be tested	Test method		Specimens		Conditioning procedure (see 4.5.1.5)	Unit of value
	ASTM	Modified by	Form and dimensions	Number tested		
Arc resistance	D 495	----	4-inch disk, 1/8 inch thick	3	A	Seconds (minimum average)
Dielectric constant at 1 megahertz	D 150	----	4-inch disk, 1/8 inch thick	3	E-48/50+D-24/23	Maximum average
Dissipation factor at 1 megahertz	D 150	----	4-inch disk, 1/8 inch thick	3	E-48/50+D-24/23	Maximum average
Dielectric strength short time	D 149	4.5.2.1	4-inch disk, 1/8 inch thick	3	E-48/50+D-48/50	Volts per mil (minimum average)
Flexural strength	D 790	4.5.2.2	5-inch bar, 1/4 by 1/2 inch	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Impact strength	D 256	----	In accordance with ASTM D 256	5	E-48/50+C-96/23/50	ft-lb/in notch (minimum average)
Water absorption	D 570	4.5.2.4	2-inch disk, 1/8 inch thick	3	E-24/100+des+D -48/50	Percent (maximum average)

Property to be tested	Value required for each type of compound															Units
	Type PT-F	Type GPT-7.5F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	Type MLT-20F	Type GLCP-30F	Type GLCP-50F	
Arc resistance	50	55	65	70	70	80	70	70	85	85	85	30	85	130	130	Seconds
Dielectric constant at 1 megahertz	3.4	3.6	3.7	3.7	4.0	3.8	4.0	4.0	3.5	3.5	3.6	4.0	3.5	4.8	5.2	----
Dissipation factor at 1 megahertz	0.03	0.03	0.03	0.03	0.03	0.02	0.03	0.03	0.009	0.012	0.009	0.007	0.009	0.04	0.04	----
Dielectric strength short time	350	360	370	375	400	400	400	400	450	450	450	320	450	480	480	Volts/mil
Flexural strength	13,000	15,000	19,000	20,000	24,000	20,000	24,000	24,000	26,600	28,500	31,400	22,000	22,500	25,000	28,000	lb/in^2
Impact strength, side	0.60	0.60	0.70	0.8	1.10	1.0	1.1	1.1	0.9	1.0	1.1	1.1	0.6	1.3	0.8	ft-lb/in
Water absorption	0.40	0.35	0.35	0.30	0.30	0.40	0.30	0.30	0.69	0.60	0.55	0.08	0.60	0.08	0.08	Percent

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4.4.1 Batch. A batch shall be considered a homogeneous unit of finished molding compound manufactured at one time.

4.4.2 Preparation of specimens. Specimens shall be prepared as specified in 4.5.1.

4.4.3 Rejection. Failure to conform to the requirements of table IV shall result in rejection of the batch of compound.

4.5 Test procedures.

4.5.1 Standard test specimens.

4.5.1.1 Number. The minimum number of standard test specimens to be tested shall be as specified in tables II, III, and IV, as applicable.

4.5.1.2 Form. The form of the standard test specimens shall be as specified in the applicable ASTM standard, and as specified in tables II, III, and IV, as applicable.

4.5.1.3 Molding of test specimens. Test specimens shall be molded by methods which are representative of commercial practice. Special treatments shall not be used to improve the properties of the specimens when compared with parts molded in commercial production.

4.5.1.4 Tolerance. Unless otherwise specified (see 6.2.1), tolerance on dimensions shall be plus or minus 5.0 percent.

4.5.1.5 Conditioning. Standard test specimens shall be conditioned before test, as specified in tables II and IV.

4.5.1.5.1 Nomenclature. The following letters shall be used to indicate the respective general conditioning procedures:

Condition A - As received; no special conditioning.
Condition C - Humidity conditioning.
Condition D - Immersion conditioning in distilled water.
Condition E - Temperature conditioning.
Condition desiccation - Desiccation condition, cooling over silica gel or calcium chloride in a desiccator at 23 degrees Celsius (°C) for 16 to 20 hours after temperature conditioning.

4.5.1.5.2 Designation. Conditioning procedures shall be designated as follows:

(a) A capital letter indicating the general condition of the specimen; that is, as received, humidity, immersion, or temperature conditioning.
(b) A number indicating in hours the duration of the conditioning.
(c) A number indicating in °C the conditioning temperature.
(d) A number indicating relative humidity (R.H.), whenever R.H. is controlled.

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The numbers shall be separated from each other by slant marks, and from the capital letter by a dash. A sequence of conditions shall be denoted by use of a plus (+) sign between successive conditions.

Examples:

Condition C-96/23/50 - Humidity condition, 96 hours at 23 ± 1.1°C and 50 ± 2 percent R.H.
Condition D-48/50 - Immersion condition, 48 hours at 50 ± 1°C.
Condition E-48/50 - Temperature condition, 48 hours at 50 ± 3°C.
Condition E-48/50+C-96/23/50 - Temperature condition, 48 hours at 50 ± 3°C, followed by humidity condition, 96 hours at 23 ± 1.1°C and 50 ± 2 percent R.H.

4.5.2 Methods of test. Each test measurement, with the exception of dielectric strength, dielectric breakdown, heat resistance, and volume and surface resistance (see 4.5.2.1, 4.5.2.1.2, 4.5.2.5, and 4.5.2.7) shall be taken at standard laboratory atmosphere of 23 ± 1.1°C and 50 ± 2 percent R.H. The test methods shall be in accordance with the applicable ASTM standard (see tables II and IV) modified as applicable.

4.5.2.1 Dielectric strength. Thermoplastic molding compounds shall be tested for dielectric strength in accordance with ASTM D 149. Tests shall be made under oil at a frequency not exceeding 100 hertz (Hz) at the temperature of the final conditioning. Specimens shall meet the requirements specified in 3.2.4.

4.5.2.1.1 Short-time test. The voltage shall be increased uniformly at the rate of 500 volts per second.

4.5.2.1.2 Dielectric breakdown. Thermoplastic molding compounds shall be tested for dielectric breakdown in accordance with ASTM D 149. The electrodes shall be brass or stainless steel cylinders 1 inch in diameter by 1 inch long with the edges rounded to a 1/8-inch radius. Tests shall be made under oil at a frequency not exceeding 100 Hz at the temperature of the final conditioning. Specimens shall meet the requirements specified in 3.2.4.

4.5.2.2 Flexural strength. Thermoplastic molding compounds shall be tested for flexural strength in accordance with ASTM D 790. The span-depth ratio shall be 16 to 1. Specimens shall meet the requirements specified in 3.2.4.

4.5.2.3 Heat-deflection temperature. Thermoplastic molding compounds shall be tested for heat-deflection temperature in accordance with ASTM D 648. Specimens shall be tested directly in an oil bath and not in air. The stress load shall be 264 pounds per square inch (lb/in^2). Specimens shall meet the requirements specified in 3.2.4.

4.5.2.4 Water absorption. Thermoplastic molding plastics shall be tested for water absorption in accordance with ASTM D 570, modified as follows:

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(a) The specimens shall be conditioned at 100 ± 2°C for 24 hours, followed by a 16 to 20 hour period of cooling over silica gel or calcium chloride in a desiccator at 23 ± 1.1°C.
(b) The specimens shall be immersed in distilled water maintained at a temperature of 50 ± 1°C for 48 hours. The information shall consist of the percentage increase in weight during immersion calculated to the nearest 0.01 percent as follows:

$$\text{Increase in weight (percent)} = \frac{\text{wet weight-conditioned weight}}{\text{conditioned weight}} \times 100$$

Specimens shall meet the requirements specified in 3.2.4. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.5.2.5 Heat resistance. Thermoplastic molding compounds shall be tested for heat resistance as specified in 4.5.2.2, modified as follows:

(a) Specimens of PT-F shall be given an additional conditioning of 1 hour at 150 ± 2°C and then be tested at 23 ± 2°C.
(b) Specimens of GPT-7.5F, GPT-15F, GPT-20F, GPT-30F, GAT-22F, GET-30F, GMT-40F, GLT-10F, GLT-20F, GLT-30F, GST-40F, MLT-20F, GLCP-30F, and GLCP-50F shall be given an additional conditioning of 1 hour at 200 ± 2°C, and then be tested at 23 ± 2°C.

The average of five such determinations divided by the average flexural strength (determined as specified in 4.5.2.2) shall be multiplied by 100, and shown as percent flexural strength retained. Specimens shall meet the requirements specified in 3.2.4. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.5.2.6 Dimensional stability (high temperature). Specimens shall be machined so the 1/2 by 1/2-inch ends are smooth and parallel. Specimens shall be subjected to the condition C-96/23/50 (see 4.5.1.5.2). The initial length of the specimens shall then be measured to the nearest 0.001 inch. The specimens shall then be subjected to 10 cycles, with each cycle consisting of the following:

(a) Specimens shall be placed in a circulating air oven at 125 ± 5°C for 48 hours.
(b) After this procedure, specimens shall be subjected to 23 ± 1.1°C and 50 ± 2 percent R.H. for 24 hours.

At the completion of 10 cycles, the final length of the specimens shall be measured to the nearest 0.001 inch. The percentage dimensional change shall be calculated to the nearest 0.1 percent as follows:

$$\text{Dimensional change (percent)} = \frac{\text{(initial length - final length)}}{\text{initial length}} \times 100$$

The average percent dimensional change of the five specimens shall be shown. Specimens shall meet the requirements specified in 3.2.4. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

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4.5.2.7 Volume and surface resistance. Thermoplastic molding compounds shall be conditioned at C-720/70/100+dew for the volume and surface resistance test as specified in 4.5.2.7.1 through 4.5.2.7.6, inclusive.

4.5.2.7.1 Specimens. Five 4-inch diameter, 1/8-inch thick specimens shall be used. Specimens shall be cleaned by noninjurious methods to prevent contamination. Precautions shall be taken in handling the specimens to avoid additional contamination. Five specimens shall be used for the measurements of this test.

4.5.2.7.2 Electrodes. Electrodes shall consist of a guarded electrode 2 inches in diameter, a 1/4-inch guard ring spaced 1/4 inch from the guarded electrode on the same side, and a third electrode 3 inches in diameter on the opposite side, and concentric with the guarded electrode. Dimensions of electrodes shall be maintained to a tolerance of plus or minus 1/64 inch. Silver paint, permeable to moisture (Dupont no. 4517, or equal) shall be used for painting electrodes on the specimens. Electrodes shall exhibit a resistance of not more than 5 ohms before and after the C-720/70/100+dew conditioning, when measured with potentials of not greater than 3 volts between points diametrically opposite on each electrode. The specimens shall be permitted to air dry after painting for at least 1 week in an atmosphere of less than 60 percent R.H. at a temperature of 25 $\pm$ 5°C.

4.5.2.7.3 Humidity chamber. The humidity chamber shall consist of a glass container with a corrosion-resistant cover. The cover shall be provided with through-panel insulators. The insulators shall serve as supports for the electrode holders as shown on figure 1. The chamber shall be of such size that the ratio of specimens' surface area to water surface area shall not exceed 2:5. The ratio of volume of air in the humidity chamber to surface area of the water shall not exceed 10:1. One hundred percent R.H. with condensation shall be obtained by natural evaporation from a quantity of distilled water located at the bottom of the chamber. The cover shall be sealed to the chamber by an inert sealing compound applied to the exterior points formed by the cover and the walls of the chamber. A small vent hole shall be provided in the cover to equalize pressure. The vent hole shall be sealed as soon as the air temperature in the humidity chamber has reached 70°C.

4.5.2.7.4 Specimen holders. Specimens shall be installed in a vertical plane in the conditioning chamber with the lower edge of the specimen not closer than 1 inch from the surface of the water. Specimens shall be held in position by electrode contactors as shown on figure 1. Electrical connection to the specimen holders shall be made by means of through-panel insulators. Insulators shall be constructed to withstand the adverse conditions within the chamber without excessive loss of insulating properties. (Insulator resistance to cover plate shall exceed 10 megohms at all times.) Polytetrafluoroethylene insulators on the humidity side of the conditioning chamber are recommended to meet this requirement. If used, these shall be cleaned with alcohol before the start of each test. Electrode contactors, and all other metallic parts of the sample, shall be silver-plated. Contact pressure against the electrodes shall be provided by backing the contactors with phosphor bronze springs or other corrosion-resistant spring material.

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4.5.2.7.5 Heating chamber. The humidity chamber shall be installed in an oven or other heating chamber constructed to maintain a temperature of 70 ± 1°C. The rate of heating of the oven shall be such that the air temperature at a point near the volumetric center of the humidity chamber shall be 70°C in 4 ± 1 hours. The quantity of water in the chamber shall be such that the water temperature shall be 65°C in 4 ± 1 hours. Room temperature shall be maintained at 25 ± 5°C. The insulation of the conductors connecting the through-panel insulators to the measuring equipment shall not be significantly deteriorated by the elevated temperatures encountered in the oven. Polytetrafluoroethylene coated wire is recommended.

4.5.2.7.6 Measurements. Volume and surface resistances shall be measured by the three-terminal method employing measuring equipment such as a megohm bridge constructed to apply 500 volts direct current (Vdc) to the specimen. A single set of measurements shall be made on each specimen while in the conditioning chamber after 30 days of the specified conditioning. Conversion of the measurements to resistivities is not required since electrode dimensions are specified. The potentials shall be applied to the specimens as shown on figure 2, or with polarities opposite to those shown on figure 2. Surface resistance measurements shall be made on the same specimens as those used for volume resistance, except that the potentials of guard and low electrodes shall be interchanged. The volume and surface resistance shall be measured, in each case, 1 minute after the potentials are applied. Low values of volume and surface resistance (below 5 mogohms) shall be measured by the circuits shown on figure 3.1/ Specimens shall meet the requirements specified in 3.2.4.

4.5.2.8 Toxicity when heated. Thermoplastic molding compounds shall be tested for toxicity in accordance with AD297457. Specimens shall meet the requirements specified in 3.2.4.

4.5.2.9 Flame resistance. Flame resistance shall be determined in accordance with method II of ASTM D 229 with the modifications specified in 4.5.2.9.1 through 4.5.2.9.7, inclusive. Specimens shall meet the requirements specified in 3.2.4.

1/ Because of the variability of the resistance of a given specimen with test conditions, and non-uniformity of the same material from specimen to specimen, determinations are usually not reproducible to closer than 10 percent, and are often even more widely divergent. A range of values of 10 to 1 may be obtained under apparently identical conditions. Errors in resistance determinations may result from the fact that the current measuring device is shunted by the resistance between the guarded terminal and the guard system. In some bridge techniques, a standard resistor in the bridge is shunted by the resistance between the unguarded terminal and the guard system. To assure validity of the volume and surface resistance measurements obtained by the bridge methods, the resistance between the unguarded and the guard terminal should be at least five times greater than the standard resistance employed in the bridge. This may be ascertained by direct two-terminal measurements between these two terminals.

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4.5.2.9.1 Weight and weight loss. Determinations of weight and weight loss, such as in definitions, apparatus, procedures, calculations, and report sections, are not applicable.

4.5.2.9.2 Flame cabinet. The 9/16-inch slot at bottom of the flame cabinet shall be on all four sides. The door shall be provided with a 1-1/4 inch diameter peep hole located directly opposite the heater coil when the door is closed. The hole shall be kept closed during testing by means of a cover.

4.5.2.9.3 Pyrometer. The means of correction from black-body radiation to actual conditions of this test shall be accomplished with a pyrometer calibrated for black-body emission. In addition, 6°C shall be added to the pyrometer to obtain the true temperature of the Nichrome V, or equal, coil.

4.5.2.9.4 Specimens. Test specimens shall be as follows:

(a) Specimens shall be molded to 1/2 by 1/2 by 5 $\pm$ 1/16 inches.
(b) The test sample shall consist of five test specimens.

4.5.2.9.5 Calibration. Calibration of this equipment shall be accomplished by adjusting the heater current to obtain an equilibrium temperature of 860 $\pm$ 2°C.

4.5.2.9.6 Calculation of burning time. Burning time shall be calculated by arranging the five values of burning time in increasing order of magnitude, as T_1, T_2, T_3, T_4, and T_5. The following ratios shall be computed:

$$\frac{T_2 - T_1}{T_5 - T_1} \quad \text{and} \quad \frac{T_5 - T_4}{T_5 - T_1}$$

If either of these ratios exceeds 0.642, then T_1 or T_5 shall be considered abnormal, and shall be eliminated. The burning time shown shall be the average of the remaining four values. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.5.2.9.7 Average ignition time. The ignition time shall be calculated as the arithmetic mean of the five specimens.

4.5.2.10 Flammability. Flammability rating shall be determined in accordance with UL 94. Results shall be shown as V-0. Specimens shall meet the requirements specified in 3.2.4. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.5.2.11 Comparative tracking index (CTI). The CTI shall be determined in accordance with IEC Publication 112. Specimens shall be as specified in 3.2.4.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

5.1 The material shall be commercially packaged in an air-tight and moisture proof bag and weigh approximately 40 - 50 pounds per package. This packaging will be used for shipment as well as in storage. Also, this package shall be sealed at all times until it is used.

6. NOTES

6.1 Intended use. The general purpose thermoplastic compounds covered by this specification are intended for applications requiring good electrical and mechanical properties. The compounds are used in electronic connectors and similar applications.

6.1.1 Flame resistant types of thermoplastic compounds. Flame retardants reduce arc tracking resistance of the compounds and should not be used in applications above 300 volts.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Color required if other than natural color (see 3.2.3).
(d) Tolerance, if other than plus or minus 5.0 percent (see 4.5.1.4).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.5.2.4, 4.5.2.5, 4.5.2.6, 4.5.2.9.6, and 4.5.2.10	Certificate of compliance	DI-E-2121	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24519 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Natural colors of molding compounds. The natural color of the various types of molding compounds approximate the following color chips of FED-STD-595.

Types	FED-STD-595	Chip number
PT-F	White	37886
GPT-7.5	White	37778
GPT-15F	White	37778
GPT-20F	White	37778
GPT-30F	White	37778
GAT-22F	Dark straw	20318
GET-30F	Light buff	13522
GMT-40F	Beige	30318
GLT-10F	Brown	20100
GLT-20F	Brown	20122
GLT-30F	Brown	20142
GST-40F	Brown	10632
GST-40F	Black	17038
MLT-20F	Greenish/brown	1/ 24098/24201
GLCP-30F	White	37778
GLCP-50F	White	37778

1/ The greenish/brown chip number falls between green chip 24098 and brown chip 24201.

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6.5 Subject term (key word) listing.

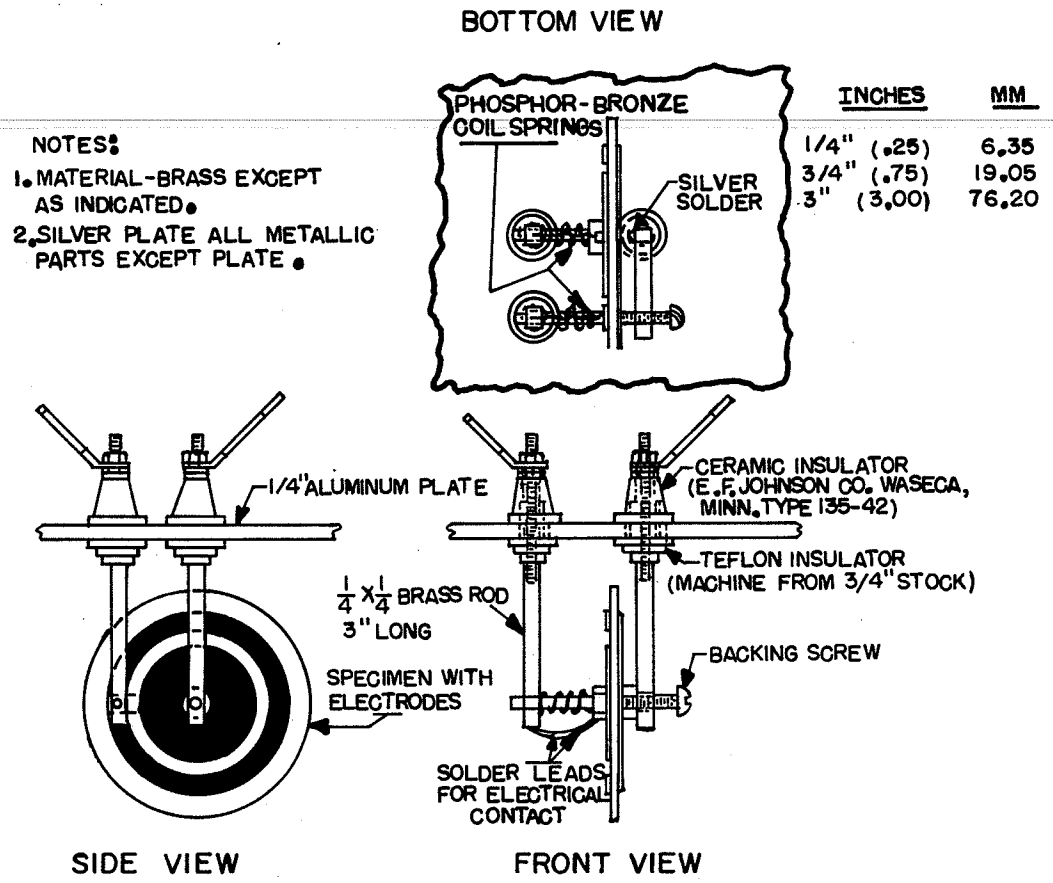
Dielectric
Flame resistant
Glass reinforced
Plastics, electrical molding
Polyarylether
Polyester
Polyetherimide
Polyphenylene

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

User activities:
Navy - AS, EC, MC

Preparing activity:
Navy - SH
(Project 5970-N665)

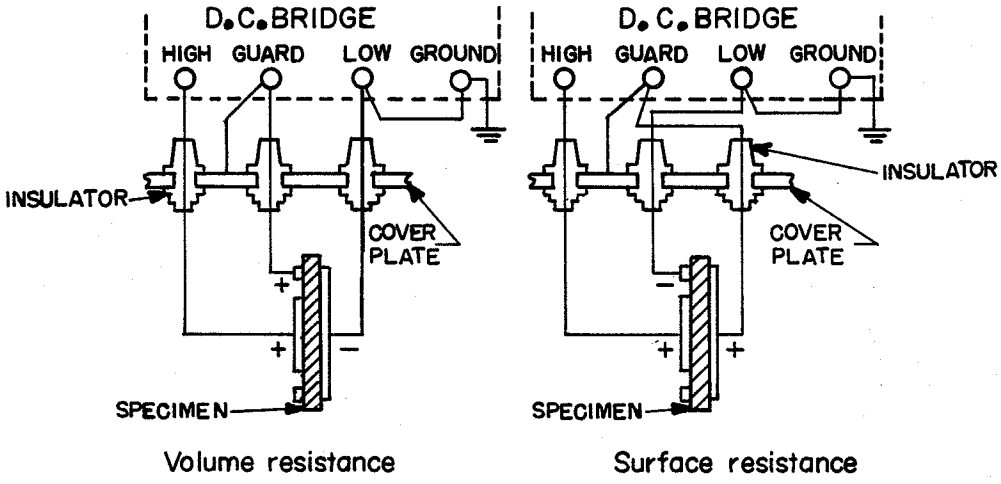
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SH 132316941

FIGURE 1. Specimen holders, electrodes, test samples, and humidity chamber cover - volume and surface resistance test.

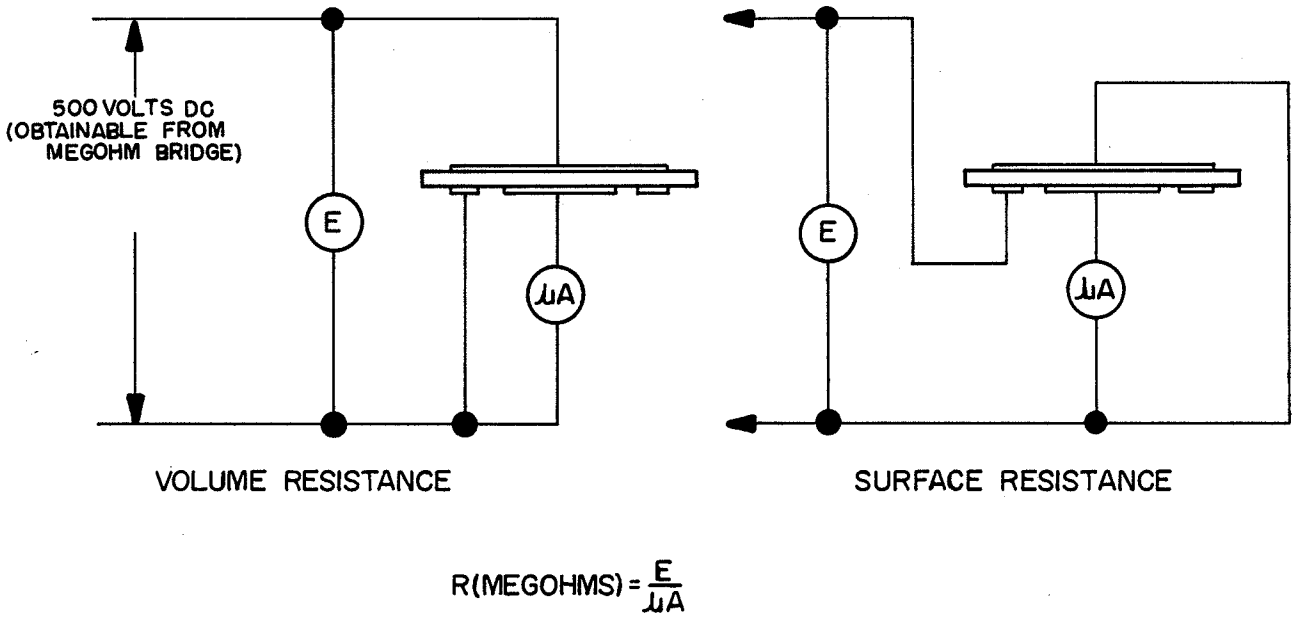
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SH 132316942

FIGURE 2. Arrangements for volume resistance and surface resistance test.

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SH 132316943

FIGURE 3. Circuits for measuring low values of volume and surface resistance.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-24519C(NAVY)	MOLDING PLASTICS, ELECTRICAL, THERMOPLASTIC

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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